Jungyu Jin

AI Research Engineer · LLM Quantization & Efficient Inference

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Research engineer working on model quantization and efficient inference for LLMs and edge models. At Nota AI I develop the company's quantization toolkit, optimize LLM, vision and speech models for CPUs, GPUs and NPUs in projects with global semiconductor companies, and write custom CUDA and Arm NEON kernels. M.S. from POSTECH (Efficient Computing Lab); co-first author of OWQ (AAAI'24 Oral) and QEFT (EMNLP'24 Findings), with 6 papers at AAAI, ACL and EMNLP.

EXPERIENCE

Nota AI — AI Research Engineer, Quantization Team — Feb. 2026 – Present

Quantization toolkit — PTQ for LLMs and edge models

- Core developer of the company's quantization toolkit, which covers LLM / VLM / speech models and edge vision models.
- Redesigned the LLM quantization pipeline and algorithm architecture, and built a parity-test harness to keep results identical during the refactor.
- Made major PTQ algorithms (GPTQ, AWQ, AutoRound, HQQ) faster and lighter: up to **~2× faster** calibration and large reductions in peak GPU memory.
- Added NVFP4 export, Mixture-of-Experts and speech-model support, and sensitivity-based mixed-precision quantization.

Hardware partner projects — model optimization for CPUs and NPUs

- Optimized LLMs and vision models for Arm CPUs and embedded NPUs, bringing quantized accuracy close to FP and above publicly available reference models.
- Improved a mobile SoC vendor's on-device quantizer (algorithm speedups of **~8×**, new mixed-precision modes, sensitivity analysis), and led a hands-on quantization seminar for its engineers.
- Reproduced an NPU's integer arithmetic on GPU to quantize and evaluate a vision-language model for that NPU, matching the vendor's accuracy.
- Ran low-bit LLM quantization experiments (NVFP4, INT3/4) and vLLM benchmarks for a data-center AI accelerator.

Research — quantization-aware speculative decoding

- Researched speculative decoding that reuses a quantized model as its own draft, with no extra memory or training.
- Wrote custom **CUDA** and **Arm NEON** kernels, and built a live on-device LLM demo on Apple Silicon for an industry exhibition.

Engineering & team

- Release owner for the quantization toolkit; maintained CI and code-quality checks.
- Introduced spec-driven, AI-agent-assisted development to the team.
- Onboarded new hires, interviewed intern candidates, and gave internal paper seminars.

Efficient Computing Lab, POSTECH — Research Intern → M.S. Student (Advisor: Eunhyeok Park) — Feb. 2023 – Aug. 2025

LLM weight quantization (OWQ, QEFT, AMQ), long-context attention scaling (SEAL), decoding factuality (PruneCD), and system design for accelerated LLM inference.

Video Intelligence Lab, Kyungpook National University — Undergraduate Researcher — Jul. 2022 – Jan. 2023

Object-ratio-preserving video retargeting with instance segmentation and video inpainting (WACV'23 Workshop).

PUBLICATIONS

* co-first author

EMNLP 2025 Main	**AMQ: Enabling AutoML for Mixed-precision Weight-Only Quantization of Large Language Models.** Sangjun Lee, Seung-taek Woo, **Jun-gyu Jin**, Changhun Lee, Eunhyeok Park.
EMNLP 2025 Main	**PruneCD: Contrasting Pruned Self Model to Improve Decoding Factuality.** Byeongho Yu, Changhun Lee, **Jun-gyu Jin**, Eunhyeok Park.
ACL 2025 Main	**SEAL: Scaling to Emphasize Attention for Long-Context Retrieval.** Changhun Lee, Minsang Seok, **Jun-gyu Jin**, Younghyun Cho, Eunhyeok Park.
EMNLP 2024 Findings	**QEFT: Quantization for Efficient Fine-Tuning of LLMs.** Changhun Lee*, **Jun-gyu Jin***, Younghyun Cho, Eunhyeok Park.
AAAI 2024 Oral	**OWQ: Outlier-Aware Weight Quantization for Efficient Fine-Tuning and Inference of Large Language Models.** Changhun Lee*, **Jun-gyu Jin***, Taesu Kim, Hyungjun Kim, Eunhyeok Park. Code: github.com/xvyaward/owq (85★)
WACV 2023 Workshop	**Object-Ratio-Preserving Video Retargeting Framework Based on Segmentation and Inpainting.** **Jun-gyu Jin***, Jaehyun Bae*, Han-gyul Baek, Sang-hyo Park.

EDUCATION

M.S., POSTECH — Sep. 2023 – Aug. 2025
Graduate School of Artificial Intelligence · Advisor: Eunhyeok Park

B.S., Kyungpook National University — Mar. 2018 – Feb. 2023
Electronics Engineering (Minor: Computer Science & Engineering) · GPA 4.04 / 4.3

AWARDS

Bronze Prize (3rd), CS & Engineering — Feb. 2024
30th Samsung Humantech Paper Award

3rd Prize — LLM inference acceleration — Nov. 2023
Samsung Computer Engineering Challenge 2023

SKILLS

Quantization	GPTQ, AWQ, SmoothQuant, AutoRound, AdaRound, QuaRot, HQQ, CLE, mixed precision (HAWQ / Fisher), NVFP4 / INT4 / INT8, MoE and VLM quantization
Inference	Speculative decoding, vLLM, llama.cpp, ExecuTorch, XNNPACK, torchao, compressed-tensors, ONNX
Kernels & HW	CUDA, Arm NEON, Metal · NVIDIA GPUs, Apple Silicon, Raspberry Pi 5, Arm Ethos-U, mobile and data-center NPUs
Languages	Python, C / C++ · PyTorch, TensorFlow